MODULE 14 · DOMAIN 3 - INTEGRATION

RAG Pipelines: Chunking, Indexing, Retrieval

The pipeline end to end, chunking by the shape of the corpus, retrieval matched to the query, keeping the index true, and grounded answers with citations.

DOMAIN 3 of 7 EXAM WEIGHT 19% MODULE 14 of 38

admin@sthithapragna.com

Most RAG failures are retrieval failures. If the right passage never arrives, no model can use it.

admin@sthithapragna.com

The pipeline, end to end

01
Ingest, clean
Extract text, strip noise, keep structure.

02
Chunk
Split where meaning breaks.

03
Enrich, embed
Add context and metadata, then vectors.

04
Index
Vector, keyword, filters. Kept fresh.

05
Retrieve, rerank
Cast wide, then rank tightly.

06
Generate, cite
Answer from sources, with citations.

ON THE EXAM

A failure at any stage reaches the answer. Diagnose by stage, earliest first.

Chunk by the shape of the corpus

CORPUS SHAPE	CHUNKING THAT FITS
Long structured documents	Split on headings and clauses; keep the section path
Short, varied records	One record per chunk, with its metadata
Source code	By function or class, with file path and signature
Deep hierarchies and specs	Retrieve small chunks, return the parent section
Tables	Keep rows with their headers, never split mid-row

ON THE EXAM

Cut where the meaning breaks, not every few hundred tokens. Standard practice, not an Anthropic rule.

Dense or sparse retrieval?

DENSE: EMBEDDINGS

Finds what you mean

- Paraphrase and synonyms
- Natural-language questions
- Concepts across wording
- Misses exact codes and rare names

SPARSE: KEYWORD, BM25

Finds what you typed

- Error codes and part numbers
- Names, IDs, exact phrases
- Cheap and transparent
- Misses paraphrase

THE PICTURE

Hybrid runs both and merges them: the librarian who understands you, plus the index that finds QX-7741.

Match retrieval to the data and the query

QUERY AND DATA	RETRIEVAL THAT FITS
Questions in plain language over prose	Dense, or hybrid
Codes, part numbers, exact names	Sparse, or hybrid
How many, total, latest, by region	SQL or an API over the source data
Scoped to a product, region or date	Metadata filters, then ranking
Mixed queries over a mixed corpus	Hybrid, plus reranking

THE TIEBREAKER

Vectors find passages. They do not count, sum or sort rows. That is a database's job.

Contextual retrieval and reranking

TECHNIQUE	REDUCTION IN TOP-20 RETRIEVAL FAILURES
Contextual embeddings	35%
Plus contextual BM25	49%
Plus reranking	67%, from 5.7% to 1.9%

ON THE EXAM

Prepend 50 to 100 tokens of chunk-specific context before embedding and indexing. Reranking costs latency.

Keeping the index true

- **Re-index on change:** triggered by publishing, not by the calendar.
- **Remove superseded chunks**, and version and date what remains.
- **Monitor freshness:** compare the index with the source of truth.
- **A stale index answers confidently and wrongly.** Nothing in the model looks broken.

> THE SIGNAL
>
> Documents changed, model and latency did not, answers went wrong: look at the index first.

The Claude pieces of a RAG system

PIECE	WHAT TO KNOW, OCTOBER 2026
Embeddings	Anthropic offers none; its docs use Voyage AI, and say to assess vendors
Citations on documents	Claude cites passages of documents you supply
Search result blocks	Your chunks, with source and title, citable like web results
Where they go	In the user turn, or returned by your search tool
Citations and structured outputs	Incompatible: choose one per request

ON THE EXAM

Citations point to passages you actually supplied, so reviewers can check every claim.

> Mereside's policy Q&A splits all 1,800 documents into fixed 500-token chunks. Exceptions often sit in the clause after the rule they modify.
>
> Advisers report answers that state a rule but miss its exception.

Which change most directly addresses this?

A Re-chunk on headings and clauses, keeping each section's title and path with its chunks

B Use fixed chunks four times larger, so more text travels together

C Retrieve fifty chunks instead of five for every question

D Switch to a larger, more expensive embedding model

◆ PAUSE AND DECIDE

Which change most directly addresses this?

A Re-chunk on headings and clauses, keeping each section's title and path with its chunks

Rule and exception stay in one unit of meaning.

B Use fixed chunks four times larger, so more text travels together

Still cuts blindly, just less often. Dilutes every match.

C Retrieve fifty chunks instead of five for every question

More noise and cost; the split clause is still split.

D Switch to a larger, more expensive embedding model

Better vectors of badly cut text.

THE SIGNAL

Structured documents, meaning split across chunks: chunk on the structure.

> Tallyhawk's support agent retrieves help articles with embeddings only. Customers often paste error codes such as SYNC-4012.
>
> For those messages, retrieval returns general articles about syncing rather than the article for that exact code.

What should Corvane recommend?

A. Move to the largest available embedding model

B. Ask customers to describe the error in their own words

C. Add keyword search alongside the embeddings, and merge the results

D. List every error code and its article in the system prompt

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Move to the largest available embedding model

Exact rare tokens are a weakness of dense retrieval in general.

B Ask customers to describe the error in their own words

Throws away the most precise clue the customer gave.

C Add keyword search alongside the embeddings, and merge the results

Hybrid: exact codes matched, meaning still understood.

D List every error code and its article in the system prompt

Thousands of codes in every request, stale on every release.

THE SIGNAL

Exact codes missed by vectors: add keyword search. Hybrid is the default for mixed queries.

> Quillon's operations managers ask questions like how many discharge summaries were signed late last week, by hospital.
>
> Today an assistant retrieves chunks of exported weekly reports and often gives wrong totals. The figures live in a reporting database.

What should Corvane recommend?

A Retrieve more report chunks so every hospital is covered

B Chunk the exported reports by hospital and by week

C Add contextual retrieval to the report chunks

D Give Claude a read-only query tool over the reporting database

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Retrieve more report chunks so every hospital is covered

Asks the model to add up fragments. Totals stay unreliable.

B Chunk the exported reports by hospital and by week

Better passages, but still passages, not a query.

C Add contextual retrieval to the report chunks

Improves finding text. This needs counting.

D Give Claude a read-only query tool over the reporting database

The database counts exactly; Claude writes the query and explains.

THE SIGNAL

Counts, totals and rankings over structured data: query the source. Do not retrieve passages.

> Tallyhawk's help centre holds articles for three product editions and two regions. Many articles exist in several near-identical versions.
>
> Customers on the Standard edition in the UK are often given steps that only work on Enterprise in the US.

Which change most directly fixes this?

A Tag chunks with edition and region, and filter retrieval to the customer's own

B Tell the model in the prompt to ignore other editions

C Merge the versions into single articles covering every edition

D Add a reranker over the top hundred and fifty chunks

◆ PAUSE AND DECIDE

Which change most directly fixes this?

A Tag chunks with edition and region, and filter retrieval to the customer's own

Wrong versions never compete. The account tells you the edition.

B Tell the model in the prompt to ignore other editions

Chunks rarely say which edition they belong to.

C Merge the versions into single articles covering every edition

Longer, harder articles; the mix-up moves inside them.

D Add a reranker over the top hundred and fifty chunks

Near-identical versions rank almost identically.

THE SIGNAL

Near-identical content for different scopes: metadata filters, set from what you already know.

> Mereside's policy chunks often read like: the maximum is raised to 95% in this case. Several products contain sentences like that.
>
> Retrieval frequently returns the right sentence from the wrong product's policy.

What should Corvane recommend?

A Use smaller chunks so each holds a single fact

B Pass only the top three chunks to keep the answer focused

C Put the whole policy library in the prompt instead

D Prepend a short context naming each chunk's document and section, before embedding and keyword indexing

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Use smaller chunks so each holds a single fact

Strips even more context. The product is still missing.

B Pass only the top three chunks to keep the answer focused

Anthropic found twenty beat five or ten. The ranking is wrong.

C Put the whole policy library in the prompt instead

Several million tokens. Far beyond the window.

D Prepend a short context naming each chunk's document and section, before embedding and keyword indexing

Contextual retrieval: the chunk now says which product it is.

THE SIGNAL

Chunks that make sense only in their document: contextual retrieval.

> For a major release, Tallyhawk rewrote 1,200 help articles. The search index was not rebuilt, and the agent began giving confident, outdated answers.
>
> The index has since been rebuilt by hand. Tallyhawk asks how to stop this happening again.

Which process change should Corvane recommend?

A Rebuild the index on the first day of every month

B Add the current release number to the system prompt

C Re-index on every publish, remove superseded chunks, and alert when the index lags the help centre

D Run a regression test on the model after each release

◆ PAUSE AND DECIDE

Which process change should Corvane recommend?

A Rebuild the index on the first day of every month

Up to a month of stale answers after every release.

B Add the current release number to the system prompt

The model still only sees the old articles.

C Re-index on every publish, remove superseded chunks, and alert when the index lags the help centre

Freshness tied to the change itself, and watched.

D Run a regression test on the model after each release

The model did not change. The index did.

THE SIGNAL

Tie re-indexing to the change, remove what it supersedes, and monitor the lag.

> Quillon's discharge summaries draw on lab results, nursing notes and medication charts. Claude currently adds free-text references such as see labs.
>
> Clinicians want to check each statement against the exact record passage behind it.

What should Corvane recommend?

A Use structured outputs with a sources field for each statement

B Pass record sections as search result blocks with citations enabled

C Instruct Claude to quote the record word for word throughout

D Have a second model add references after the summary is written

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Use structured outputs with a sources field for each statement

Model-written references, unchecked; and incompatible with citations.

B. Pass record sections as search result blocks with citations enabled

Each claim links to a passage that was actually supplied.

C. Instruct Claude to quote the record word for word throughout

An unreadable summary, with no link to where quotes came from.

D. Have a second model add references after the summary is written

Matching after the fact can attach a wrong source convincingly.

THE SIGNAL

Verifiable grounding: supply sources as citable blocks and let the API link each claim.

> Brightwater Robotics wants an engineering assistant that answers questions over its 40,000-file codebase. Retrieval uses fixed 300-token chunks and a general-purpose embedding model.
>
> Retrieved chunks are often fragments of functions, cut off from their names.

Which two changes would most improve retrieval? Choose two.

A Keep fixed chunks, but overlap them by half

B Move the assistant to the most capable model tier

C Chunk by function and class, keeping the file path and signature

D Use an embedding model built for code retrieval

E Strip comments from the code before indexing

◆ PAUSE AND DECIDE

Which two changes would most improve retrieval? Choose two.

A Keep fixed chunks, but overlap them by half

Twice the index, still cut blindly.

B Move the assistant to the most capable model tier

Generation, not retrieval. Fragments still arrive.

C Chunk by function and class, keeping the file path and signature

Units of meaning in code, with their names attached.

D Use an embedding model built for code retrieval

Vectors that understand code, as code.

E Strip comments from the code before indexing

Removes the natural language that matches questions.

THE SIGNAL

Code is a corpus shape of its own: chunk by its structure, embed with a model made for it.

In five lines

- **Most RAG failures are retrieval failures.** Diagnose the pipeline by stage.
- **Chunk by corpus shape:** clauses, records, functions, parents. Not every N tokens.
- **Match the method:** dense for meaning, sparse for codes, hybrid for both, SQL for counts.
- **Contextual retrieval and reranking** cut failures sharply; filters scope by metadata.
- **Keep it true and checkable:** re-index on change, cite supplied passages.

COMING UP

Next: identity, authorisation and the security gaps that integration opens.